

Expt. No. 6(a)	Staircase Wiring

AIM:

To control the status of the given lamp using two-way switches.

APPARATUS REQUIRED:

S.NO	APPARATUS	RANGE / TYPE	QUANTITY
1	Incandescent Lamp	60W/40W	1
2	SPDT (Single Pole Double Throw-Double Way Switch)	5A, 230V	2
3	Lamp holder	Pendant type	1
4	Line Tester	500V / Taparia 813	1
5	3 Pin Plug	5A, 230V	1
6	Wire Stripper Cum Cutter	Pye 950	1
7	Connecting Wires	12A / 1 sq.mm Area/ 0.7 Thickness	As Per Requirement

THEORY:

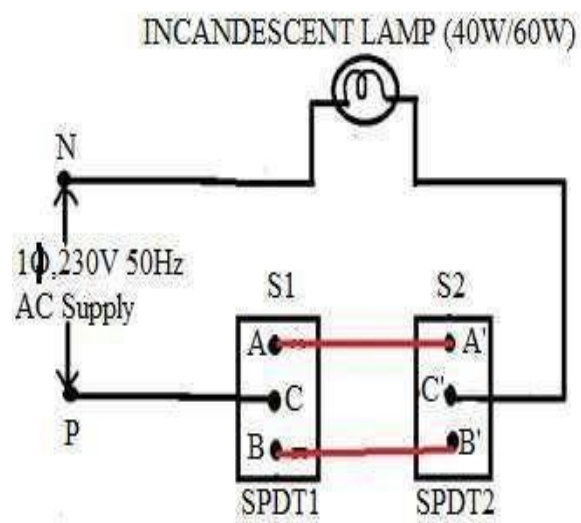
Staircase wiring is a common multi-way switching or two-way light switching connection; one light two switches wiring. One lamp is controlled by two switches from two different positions that is to operate the load from separate positions such as above or below the staircase, from inside or outside of a room, or as a two-way bed switch. The main purpose of two-way switching connection is to connect and control AC appliances and equipments from two separate locations. It is mostly used in staircase wiring where a light bulb can be control (Switch ON / Switch OFF) from different places, no matter you are in the upper or lower portion of stair; it does not depend on the switches position as well. You just have to press the switch button to OFF/ON to perform the switching operation. It is also used in rooms having large area which has two entry and exit gates. It is used to control any electrical (AC or DC) appliance or equipment like fan, light bulbs etc from two different places.

PROCEDURE:

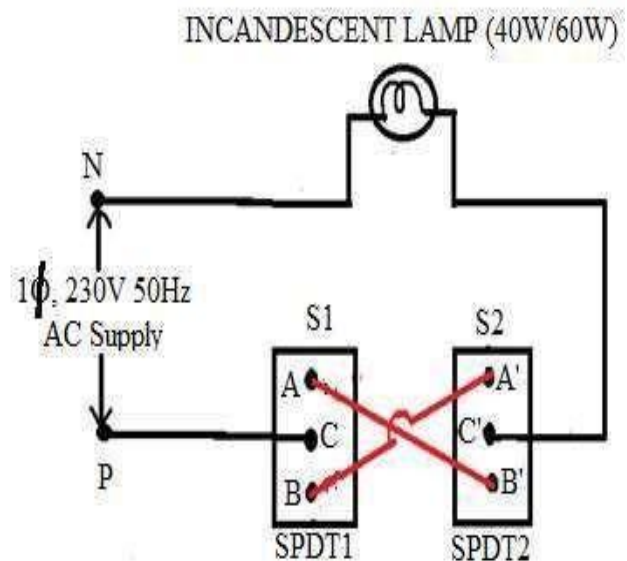
- ☐ A piece of wire is connected to the phase side and other end to the middle point of SPDT switch 1.
- ☐ Another point of lamp holder is connected to neutral line.
- ☐ Upper point of SPDT switch 1 is connected to the upper point of SPDT switch 2.
- ☐ Lower point of SPDT switch 1 is connected to the lower point of switch 2.
- ☐ Circuit is tested that all combination of switch connection.

CIRCUIT DIAGRAM:

DIRECT CONNECTION:



INDIRECT CONNECTION:



TABULATION:

DIRECT CONNECTION:

S.NO	S1	S2	LAMP STATUS
1	CA	C'A'	ON
2	CB	C'B'	ON
3	CA	C'B'	OFF
4	CB	C'A'	OFF

INDIRECT CONNECTION:

S.NO	S1	S2	LAMP STATUS
1	CA	C'A'	OFF
2	CB	C'B'	OFF
3	CA	C'B'	ON
4	CB	C'A'	ON

Marks Obtained:

Theoretical Calculations	20	
Observation	20	
Execution of practice examples	30	
Viva	10	
Record	20	
Total Score	100	
Date of experiment		Faculty signature
Date of record submission		

RESULT:

Thus the status of the given lamp was controlled and tested under direct and indirect connection using two way switches.

Expt. No. 6(b)	Fluorescent tube wiring

AIM:

To prepare wiring for a fluorescent tube light with switch control.

APPARATUS REQUIRED:

Sl No.	Apparatus	Range	Quantity
1.	Tube light with fitting	-	1
2.	Joint clips	-	As required
3.	Switch	-	1
4.	Wires	-	As required
5.	Screws	-	As required
6.	Switch board	-	1

THEORY:

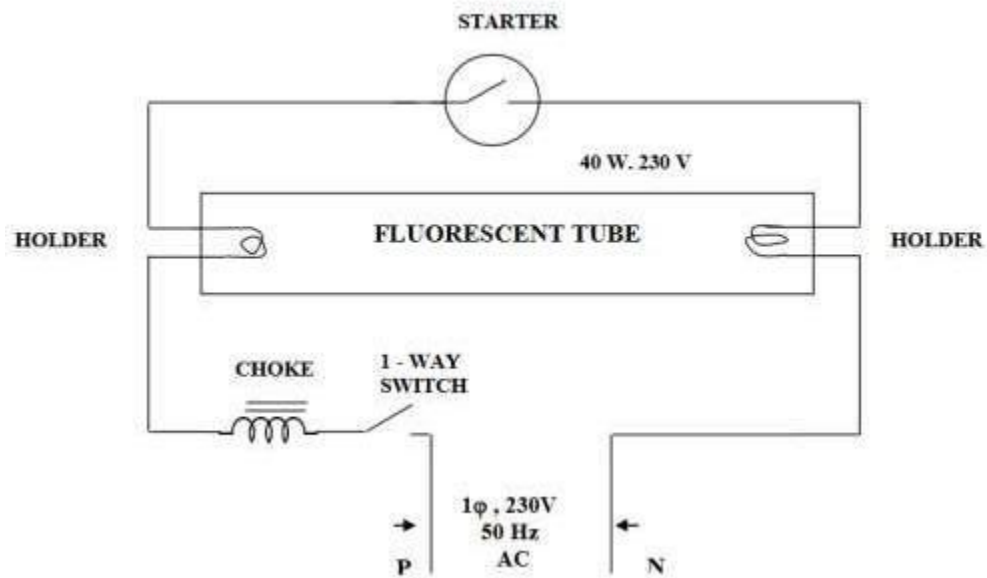
Working of the Fluorescent Tube Light:

The fluorescent lamp circuit consists of a choke, a starter, a fluorescent tube and a frame. The length of the commonly used fluorescent tube is 100 cm; its power rating is 40 W and 230V. The tube is filled with argon and a drop of mercury. When the supply is switched on, the current heats the filaments and initiates emission of electrons. After one or two seconds, the starter circuit opens and makes the choke to induce a momentary high voltage surge across the two filaments. Ionization takes place through argon and produces bright light.

PROCEDURE:

1. Mark the switch and tube light location points and draw lines for wiring on the wooden board.
2. Place wires along the lines and fix them with the help of clips.
3. Fix the switch and tube light fitting in the marked positions.
4. Complete the wiring as per the wiring diagram.
5. Test the working of the tube light by giving electric supply to the Circuit.

CIRCUIT DIAGRAM - TUBE LIGHT



Marks Obtained:

Theoretical Calculations	20	
Observation	20	
Execution of practice examples	30	
Viva	10	
Record	20	
Total Score	100	
Date of experiment		Faculty signature
Date of record submission		

RESULT:

Thus the wiring for the tube light is completed and tested.

Expt. No. 7	Output Characteristics of LVDT

AIM:

To plot the output characteristics of LVDT.

APPARATUS REQUIRED:

S.No.	Apparatus	Quantity
1	LVDT kit	1
2	Multimeter	1
3	Connecting Wires	As required

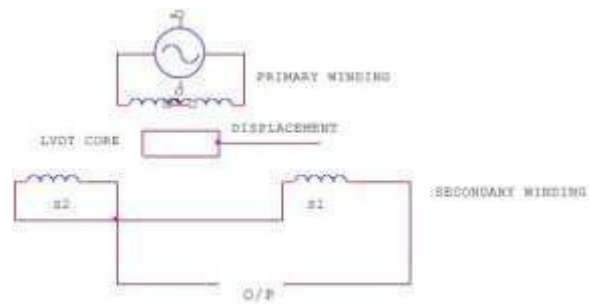
PROCEDURE:

1. Connect the circuit according to circuit diagram.
2. Switch on the power supply.
3. The core is initially brought to null position.
4. First turn the nut in clockwise direction to move core inwards i.e. left of null position & take respective voltage readings on the voltmeter.
5. Now turn nut in anticlockwise direction to move the core towards right of null point & again take respective voltage reading from voltmeter.
6. Plot the graph from the observations taken.

Marks Obtained:

Theoretical Calculations	20	
Observation	20	
Execution of practice examples	30	
Viva	10	
Record	20	
Total Score	100	
Date of experiment		Faculty signature
Date of record submission		

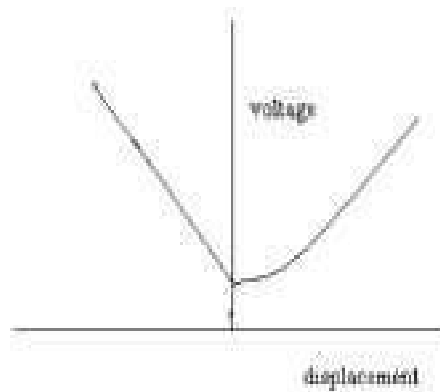
CIRCUIT DIAGRAM:



TABULAR COLUMN:

S.N.	Core Movement towards Right		Core Movement towards Left	
	Displacement (mm)	Output Voltage (mV)	Displacement (mm)	Output Voltage (mV)
1.	0	0.4	0	0.4
2.	1	1.5	-1	0.7
3.	2	2.8	-2	1.8
4.	3	4.0	-3	3.0
5.	4	5.0	-4	4.3
6.	5	6.2	-5	5.5
7.	6	7.4	-6	6.7

MODEL GRAPH:



RESULT: Thus the output characteristics of LVDT is plotted.